

1. Smart Home Automation System

Simple Reflex Agent:-

- Works using IF-THEN rules

Example: If motion is detected - Light on

Model Based Agent:-

- Remembers previous information

Example: Whether someone is already in the room

Goal Based Agent:-

- Works to achieve a goal

Example: keep room temperature at 24°C

Utility Based Agent:-

- Chooses the best action

Example:

Saves electricity while keeping the room comfortable

Best agent

Model Based Agent + Utility Agent
is the Best

Because
it learns user habits
Saves energy.
improves security

2.

Robot in an Unknown Maze

Uniform cost search

- Expands the path with the lowest cost first
- Finds the shortest cost path
- Uses more memory

Iterative Deepening Search (IDS)

- Searches one level at a time.
- Uses less memory
- Good when the maze depth is unknown

Intelligent Agents are

- Simple Reflex
- Model Based
- Goal Based
- Utility Based

Intelligent Agent

Simple Reflect:

Reacts only to the current input.

Model Based:

Remembered visited path.

Goal Based:

Try to reach the exit.

Utility Based:

It will find the shortest path and the safest one.

Best:

Goal Based + Model Based

~~the~~ Best one.

because
it remembers the maze and
find the best path for exists.

Justification

Model-Based Agent remembers explored paths.

Goal-Based Agent focuses on reaching the exit

VCS guarantees the lowest-cost path
when movement cost vary.
If all movement costs are equal, then
Based Agent + IDS is more
memory efficient and still
guarantees finding a solution

